Speech Tech Week 4

1) In K-means, the number of clusters (K) changes as the algorithm reaches to convergence. *1 point*

- True
- False ✓

it is predefined how many clusters will be there.

2) Which of the following statement is false w.r.t K-means *1 point*

- K-Means is a hard clustering algorithm ✓
- K-Means use euclidean distance to compute the distance between centroids and data points. ✓
- K-Means is a supervised learning problem ✓ → unsupervised
- None of these

3) In the Linde-Buzo-Gray(LBG) Algorithm, the number of cluster changes as the algorithm reaches convergence *1 point*

- True ✓
- False

1) State true or false: Speech is sequential data. *1 point*

- True ✓
- False

2) If there are two possible weather conditions namely "rainy" and "not rainy" on each day, how many possible combinations of weather conditions are there for the next three days? *1 point*

- 3
- 2
- 8 ✓
- None of the above

$N^T \rightarrow 2^3 = 8$

$N \rightarrow$ no. of states

$T \rightarrow$ total time period

1) If there are N states in a Markov model, how many elements will be there in the transition probability matrix? *1 point*

- N
- N^2 ✓
- $N - 1$
- $\frac{N}{2}$

$[\ \ ]_{N \times N}$

2) If there are two possible weather conditions namely "rainy" and "not rainy" on each day, how many possible combinations of weather conditions are there for the next three days? *1 point*

- 3
- 2
- 8 ✓
- None of the Above

1) For determining the best possible sequence of events in a Markov chain **Greedy Approach** is used. *1 point*

- True
- False ✓

at each step it might give best probab., not entire seq. will have best probab.

2) Given a Markov chain with 4 states, determine the total number of paths for a 6 day forecasting problem. *1 point*

- 4096 ✓
- 24
- 10
- 1296

$4^6 = 4096$

$\gamma_A(B)$ → time step B
→ reach state A — max prob. to reach state A at time step B.

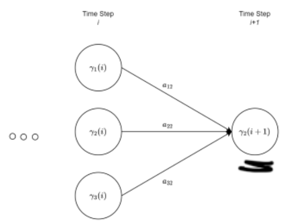
1) The notation $\gamma_A(B)$ is defined as one of the following **1 point**

- ◯ Minimum probability of reaching state A at time step B
- ◯ Average probability of reaching state B at time step A
- ◉ Maximum probability of reaching state A at time step B
- ◯ Maximum probability of reaching state B at time step A

2) a_{23} → transition from state $2 \to 3$ **1 point**

- ◯ Transition probability to state 3 at time step 2.
- ◯ Transition probability from state $3 \longrightarrow$ state 2.
- ◯ Transition probability to state 2 at time step 3.
- ◉ Transition probability from state $2 \longrightarrow$ state 3. \end{choices}

1) Consider a Markov chain with three states $(1, 2, 3)$. Fig. shows a part calculation for the viterbi algorithm. Determine the correct expression for $\gamma_2(i+1)$ **1 point**



$max(\gamma_1(i)\, a_{11},\ \gamma_2(i)\, a_{23},\ \gamma_3(i)\, a_{32})$

- ◯ $min(\gamma_1(i)a_{12}, \gamma_2(i)a_{22}, \gamma_3(i)a_{32})$
- ◉ $max(\gamma_1(i)a_{12}, \gamma_2(i)a_{22}, \gamma_3(i)a_{32})$
- ◯ $\gamma_1(i)a_{12} \times \gamma_2(i)a_{22} \times \gamma_3(i)a_{32}$
- ◯ $\gamma_1(i)a_{12} + \gamma_2(i)a_{22} + \gamma_3(i)a_{32}$

2) Let us consider an example Markov chain with two states A & B. The state transition matrix defines probability of next state from the current state as follows **1 point**

$\begin{pmatrix} P(A|A) & P(B|A) \\ P(A|B) & P(B|B) \end{pmatrix}.$

The state transition matrix for this example is given by:

$\begin{pmatrix} 0.6 & 0.4 \\ 0.37 & 0.63 \end{pmatrix}.$

Assume that the initial state is A. Determine the value of $\gamma_A(1)$

(A|A)

1 2

$\gamma_A(1) \xrightarrow{0.6} \gamma_A(2)$

0.2 0.37

$\gamma_B(1) \to \gamma_B(2)$

0.4

A

B|A

- ◯ 0.63
- ◯ 0.4
- ◯ 0.37
- ◉ 0.6

1) Let us consider an example Markov chain with two states A & B. The state transition matrix defines probability of next state from the current state as follows **1 point**

$\begin{pmatrix} P(A|A) & P(B|A) \\ P(A|B) & P(B|B) \end{pmatrix}.$

The state transition matrix for this example is given by:

$\begin{pmatrix} 0.6 & 0.4 \\ 0.37 & 0.63 \end{pmatrix}.$

$\gamma_A(2) = [0.6 \times 0.6,\ 0.4 \times 0.37] \to 0.36$

Assume that the initial state is A. Determine the value of $\gamma_A(2)$

- ◉ 0.36
- ◯ 0.148
- ◯ 0.0532
- ◯ 0.508

2) The notation $\gamma_A(B)$ **1 point**

- ◯ Minimum probability of reaching state A at time step B
- ◯ Average probability of reaching state B at time step A
- ◉ Maximum probability of reaching state A at time step B
- ◯ Maximum probability of reaching state B

→ unsupervised

3) K-Means is a _______ learning problem. *1 point*

- Self-Supervised
- Supervised
- Unsupervised
- Quasi-Supervised

4) If the next time step depends only on the current time step, then it is called ______ *1 point*

- First order Markov model → only on current time step
- Second order Markov model
- Zero order Markov model
- None of the above

5) Which of the following are examples of sequential data? *1 point*

- Weather
- Stock market
- Stock market
- All the above

6) In GMM, each point can only belong to one cluster, whereas in K-Means, each point can belong to multiple mixtures, each with a respective probability *1 point*

- True
- False

1) The speech sample $s[n], s[n] \in \mathbb{R}^D$ represents the D dimensional speech sample at the n^{th} time step, usually D is very large. To reduce bandwidth requirement an estimate $(\hat{s}[n])$ of $s[n]$ is derived at the receiver side with coefficient vector $a_k; k \in [1, ..., K]$ such that $K << D$. Following represents the correct expression of $\hat{s}[n]$: *1 point*

- $\sum_K^{k=1} a_k^{s[n-k]}$
- $\prod_K^{k=1} a_k \times s[n-k]$
- $\sum_K^{k=1} a_k \times s[n-k]$
- $\sum_K^{k=1} s[n-k]^{a_k}$

we want linear predic^n
instead of sending full s[n], we predict it using past samples.
we use wt sum of past samples

$$\underbrace{\hat{s}[n]}_{\text{predicted speech vector}} = \sum_{k=1}^{K} \underbrace{a_k}_{\text{scalar coefficient}} \times \underbrace{s[n-k]}_{\text{past vector}}$$

Yes, the answer is correct.
Score: 1

Accepted Answers:
$\sum_K^{k=1} a_k \times s[n-k]$

2) The speech sample $s[n], s[n] \in \mathbb{R}^D$ represents the D dimensional speech sample at the n^{th} time step, usually D is very large. To reduce bandwidth requirement an estimate $(\hat{s}[n])$ of $s[n]$ is derived at the receiver side with coefficient vector $a_k; k \in [1, ..., K]$ such that $K << D$. The optimal values for a_k at the receiver side could be found using the one of the following ways: *1 point*

- $min \sum (s[n] - \hat{s}[n])^2$
- $min \sum |s[n] - \hat{s}[n]|$
- $max \sum (s[n] - \hat{s}[n])^2$
- Both A & B

we are predicting s[n] using past samples & coeff. a_k.
we have to choose a_k should minimize the diff. true samples s[n] & predicted $\hat{s}[n]$

we can use MSE & MAE

Yes, the answer is correct.
Score: 1

Accepted Answers:

3) Which of the following statement is true w.r.t Linde-Buzo-Gray(LBG) Algorithm? **1 point**

- Linde-Buzo-Gray(LBG) Algorithm starts with multiple randomly selected centroids from the data points.
- Linde-Buzo-Gray(LBG) Algorithm does not converge to local maxima or global maxima
- Given a centroid (c), we get a new set of centroids by performing a small distortion of c by δ
- None of these

Yes, the answer is correct.
Score: 1

Accepted Answers:

Given a centroid (c), we get a new set of centroids by performing a small distortion of c by δ

4) Which of the following is correct according to Bayes rule? **1 point**

- $P(A|B) = \frac{P(A)}{P(B)}$
- $P(A|B) = \frac{P(A \cap B)}{P(A)}$
- $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- $P(A|B) = \frac{P(A \cup B)}{P(B)}$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Yes, the answer is correct.
Score: 1

Accepted Answers:

5) Assume that we have observed the following weather sequence: sunny, cloudy, rainy, rainy, cloudy, sunny, sunny, rainy, sunny, cloudy, cloudy, rainy, sunny. What is the probability of today being cloudy given that yesterday was cloudy? **1 point**

- 0.25
- 0.5
- 0.125
- None of these

total cloudy = 4

together = 1

$\frac{1}{4} = 0.25$

Yes, the answer is correct.
Score: 1

Accepted Answers:

0.25

6) Given the following transition probability matrix and that yesterday was cloudy, what is the probability of observing the following weather sequence: sunny, cloudy, rainy? **1 point**

$C \rightarrow S \rightarrow C \rightarrow R$

$$\begin{array}{cc} & \text{Today} \\ & \begin{array}{ccc} \text{sunny} & \text{cloudy} & \text{rainy} \end{array} \\ \text{Yesterday} \begin{array}{r} \text{sunny} \\ \text{cloudy} \\ \text{rainy} \end{array} & \begin{bmatrix} 0.25 & 0.5 & 0.25 \\ 0.25 & 0.25 & 0.5 \\ 0.5 & 0.25 & 0.25 \end{bmatrix} \end{array}$$

$0.25 \times 0.5 \times 0.5$

0.25×0.25

0.0625

- 0.25
- 0.0625
- 0.5
- 1

7) Let us consider an example Markov chain with two states A & B. The state transition matrix defines probability of next state from the current state as follows **1 point**

$$\begin{pmatrix} P(A|A) & P(B|A) \\ P(A|B) & P(B|B) \end{pmatrix}.$$

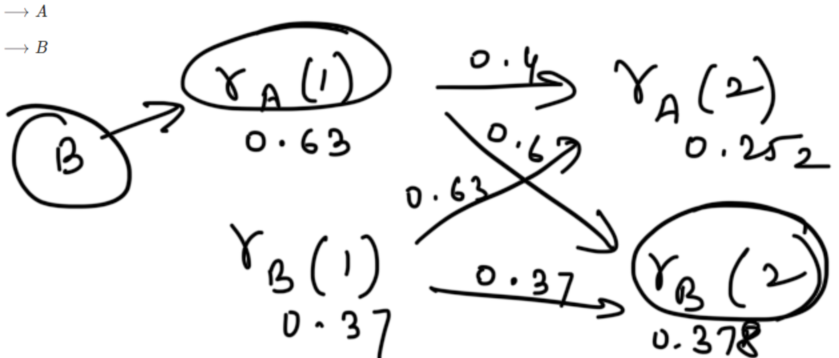
The state transition matrix for this example is given by:

$$\begin{pmatrix} 0.4 & 0.6 \\ 0.63 & 0.37 \end{pmatrix}.$$

$0.63 \times 0.4 = 0.252$

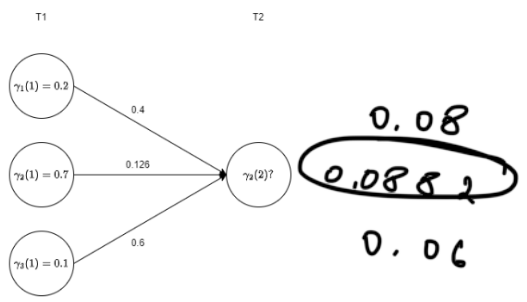
Suppose the initial state is B, what is the most probable sequence for a two day forecast?

- $B_{initial} \longrightarrow A \longrightarrow A$
- $B_{initial} \longrightarrow A \longrightarrow B$
- $B_{initial} \longrightarrow B \longrightarrow A$
- $B_{initial} \longrightarrow B \longrightarrow B$



8) Consider the Fig, and determine $\gamma_2(2)$

1 point



- 0.08
- 0.0882
- 0.006
- 0.2282